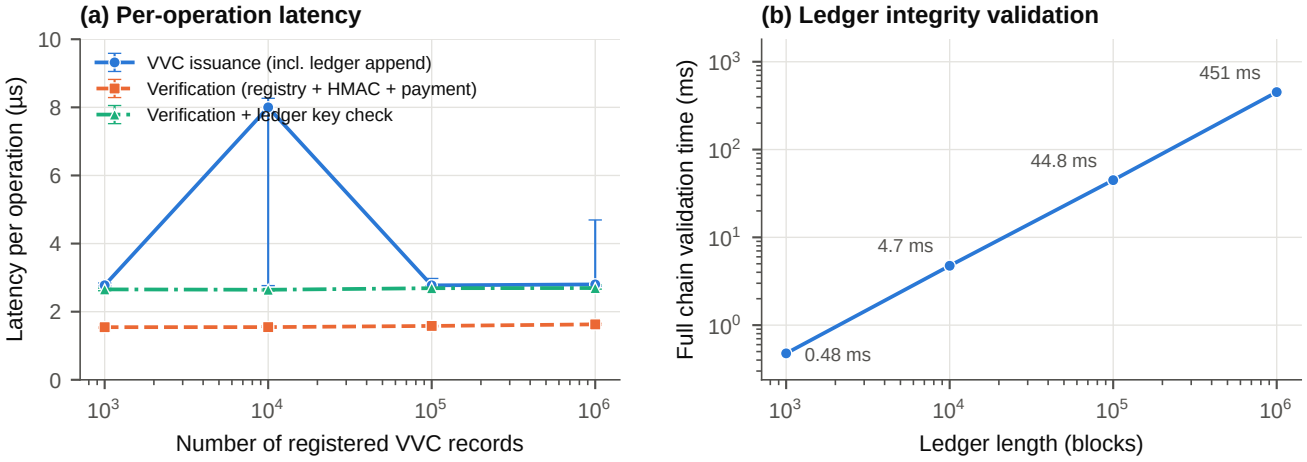
(a) Per-operation latency
Latency per operation (µs)
Number of registered VVC records
VVC issuance (incl. ledger append)
Verification (registry + HMAC + payment)
Verification + ledger key check
0
2
4
6
8
10
10^3
10^4
10^5
10^6
(b) Ledger integrity validation
Full chain validation time (ms)
Ledger length (blocks)
10^0
10^1
10^2
10^3
0.48 ms
4.7 ms
44.8 ms
451 ms